A244594 concise calculus certificate

VERIFIED

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Definition and first terms

$(4 - 1\,x)\,A(x) = 3 + A(x)^3$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(1)*T(x); T=x+1*x*T+3*T^2+1*T^3.`

$a(0), \ldots, a(7) = 1, 1, 4, 29, 263, 2672, 29088, 331749.$

Kernel used throughout: $\rho(u) = u\,\dfrac{1 - 3\,u^1 - 1\,u^2}{1 + 1\,u}$, so $x = \rho(T(x))$.

Typogeometry

`1x1` `{1,1}x4` `{1,{1,1}}x16` `{1,1,1}x1`

colored arity geometry; node color distinguishes constructors. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2 :\ 3,\ \Delta_2^{\text{marked}} :\ 1,\ \Delta_3 :\ 1.$

Coefficient and contour extraction

$$\mathcal{K}_{n,h} = \{(k_1, \ldots, k_2) \in \mathbb{Z}^2_{\geq 0} : h + \sum_{j=1}^{2} j k_j = n - 1\}, \quad K = \sum_{j=1}^{2} k_j,$$

$$a(n) = \frac{1}{n} \sum_{h=0}^{n-1} \binom{n}{h} \sum_{\boldsymbol{k} \in \mathcal{K}_{n,h}} \binom{n+K-1}{K} \binom{K}{k_1, \ldots, k_2} 3^{k_1}\, 1^{k_2}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{4} P_r(n) a(n+r) = 0$, $n \geq 1$, with

$$P_0(n) = 4\,n^3 + 2\,n^2 - 2\,n$$
$$P_1(n) = -64\,n^3 - 168\,n^2 - 136\,n - 32$$
$$P_2(n) = 384\,n^3 + 1824\,n^2 + 2848\,n + 1472$$
$$P_3(n) = -781\,n^3 - 5825\,n^2 - 14286\,n - 11520$$
$$P_4(n) = 52\,n^3 + 468\,n^2 + 1352\,n + 1248$$

Differential equation

$$(0)\,A(x) + \left(4\,x^5 - 32\,x^4 + 64\,x^3\right) A'(x)$$
$$+ \left(14\,x^6 - 168\,x^5 + 672\,x^4 - 1139\,x^3\right) A^{(2)}(x) + \left(4\,x^7 - 64\,x^6 + 384\,x^5 - 781\,x^4 + 52\,x^3\right) A^{(3)}(x) = 0.$$

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$R(n, u, x) = N/D$ (exact 246-character numerator in embedded payload).

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 6×6; remainder 2×3, rank 2, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.